20 Regression Models, 20 Different Graphs

Diagnostic plots and comparison dashboard

Dataset: scikit-learn Diabetes (442 samples, 10 features) | 80/20 train-test split (353 / 89) | random_state = 42

Overview

Each model is fitted on the same preprocessed data and gets its **own diagnostic plot** showing something specific to how that model works (e.g. coefficient shrinkage for Ridge, the tree structure for Decision Tree, learning curves for boosting models). A comparison dashboard at the end ranks all 20 models.

Preprocessing

Numeric columns are median-imputed and standardised; categorical columns (none in this dataset) would be mode-imputed and one-hot encoded. The target is `target` (disease progression one year after baseline). The data are split 80/20; training set shape (353, 10), test set shape (89, 10). Metrics reported are RMSE, MAE and R^2 on the held-out test set, plus model fit time.

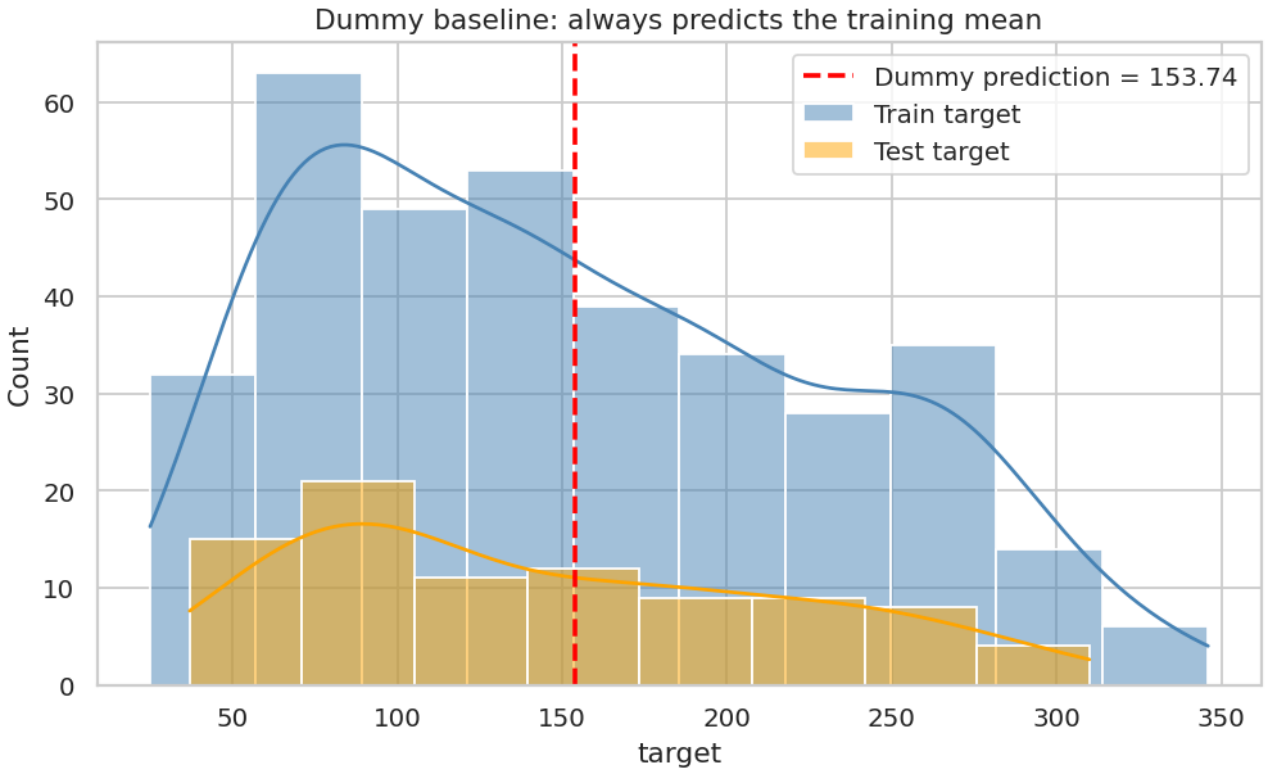
Test-set metrics at a glance (in notebook order)

#	Model	RMSE	MAE	R^2
1	Dummy mean baseline	73.22	64.01	-0.012
2	Linear Regression	53.85	42.79	0.453
3	Ridge	53.77	42.81	0.454
4	Lasso	52.92	42.79	0.471
5	ElasticNet	53.45	42.98	0.461
6	Bayesian Ridge	53.59	42.87	0.458
7	Huber Regressor	54.14	42.88	0.447
8	SGD Regressor	54.41	43.28	0.441
9	Decision Tree	57.99	46.77	0.365
10	Random Forest	54.92	44.75	0.431
11	Extra Trees	53.05	43.86	0.469
12	Gradient Boosting	53.39	43.62	0.462
13	Hist Gradient Boosting	57.87	46.09	0.368
14	AdaBoost	54.11	43.76	0.447
15	K-Nearest Neighbors	55.30	45.66	0.423
16	Support Vector Regression	50.10	39.10	0.526
17	MLP Neural Network	60.47	48.16	0.310
18	XGBoost	56.85	45.49	0.390
19	LightGBM	58.09	47.75	0.363
20	CatBoost	52.58	42.61	0.478

1) Dummy mean baseline

Diagnostic: target distribution vs constant prediction

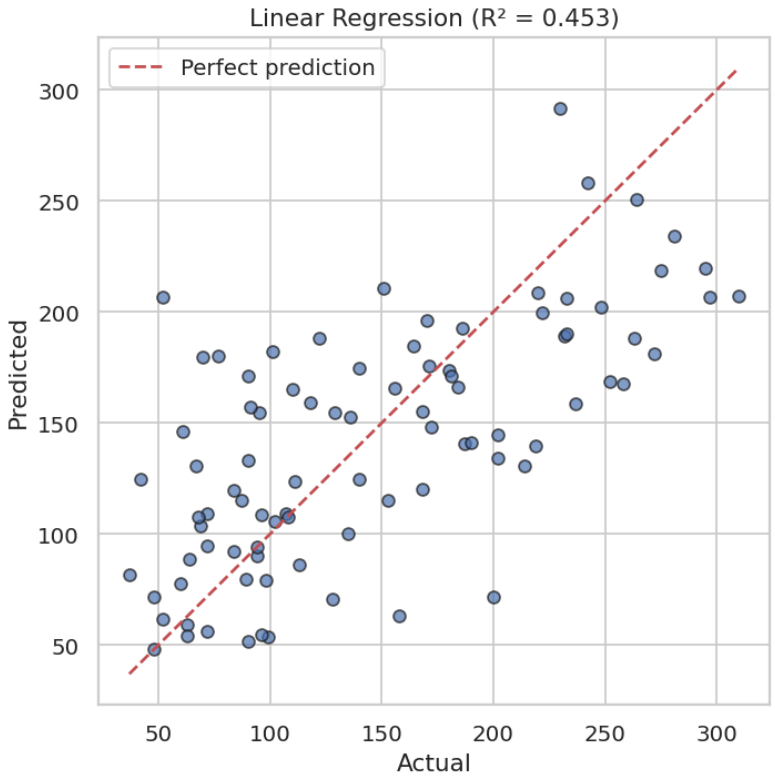
RMSE	MAE	R^2	Fit time (s)
73.22	64.01	-0.012	—



2) Linear Regression

Diagnostic: actual vs predicted

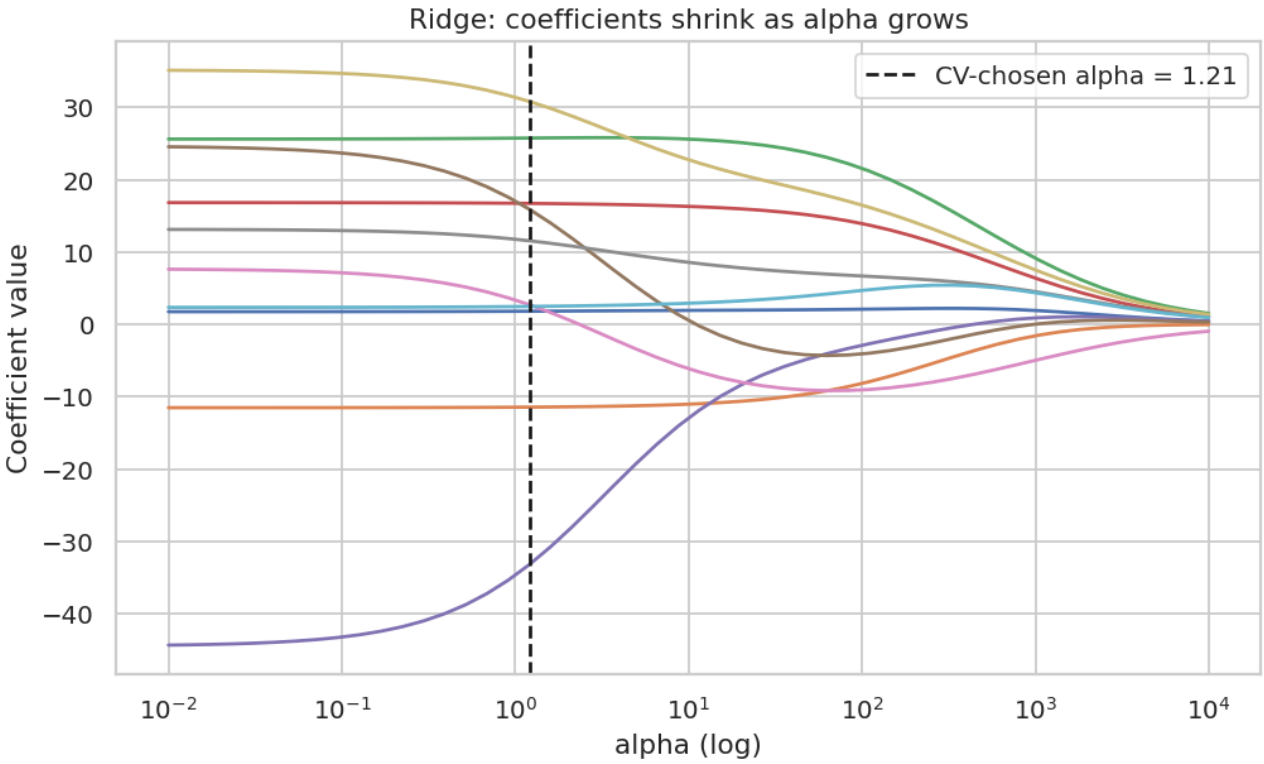
RMSE	MAE	R^2	Fit time (s)
53.85	42.79	0.453	0.123



3) Ridge

Diagnostic: coefficient shrinkage path

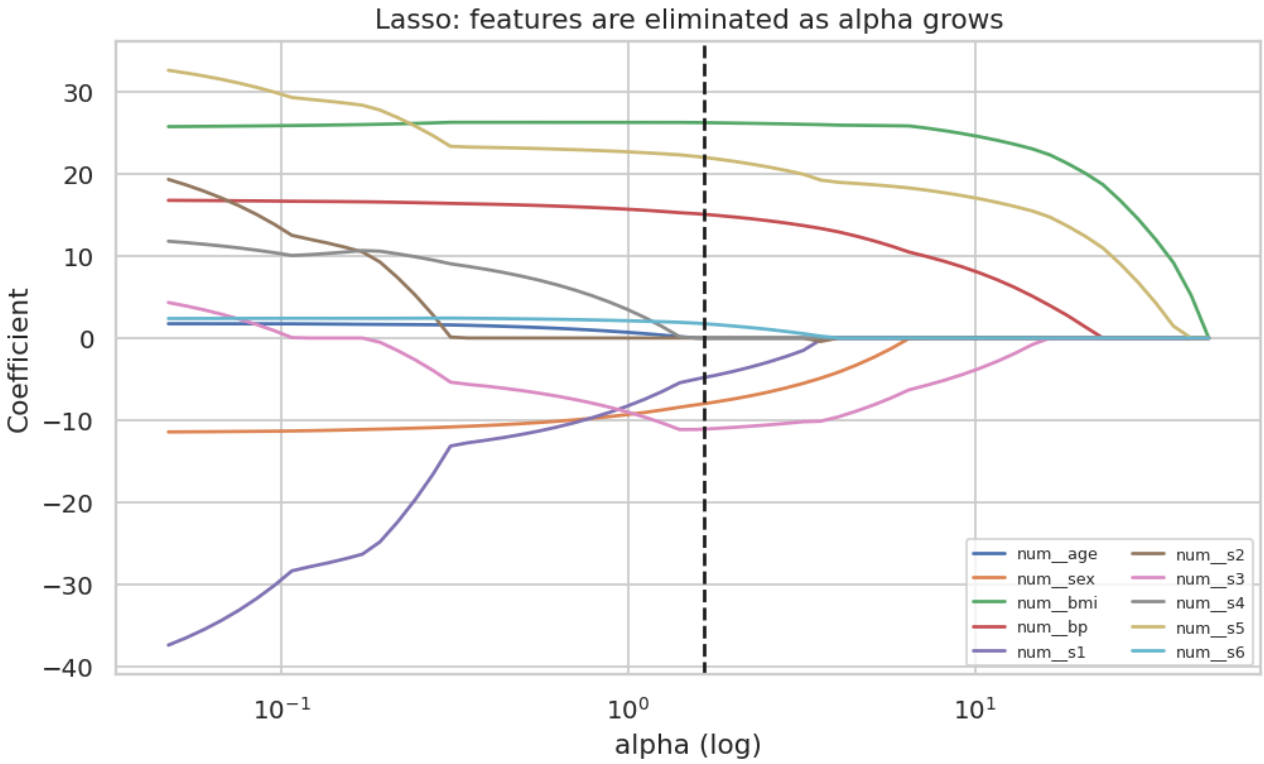
RMSE	MAE	R^2	Fit time (s)
53.77	42.81	0.454	0.007



4) Lasso

Diagnostic: coefficient path (features dropping to zero)

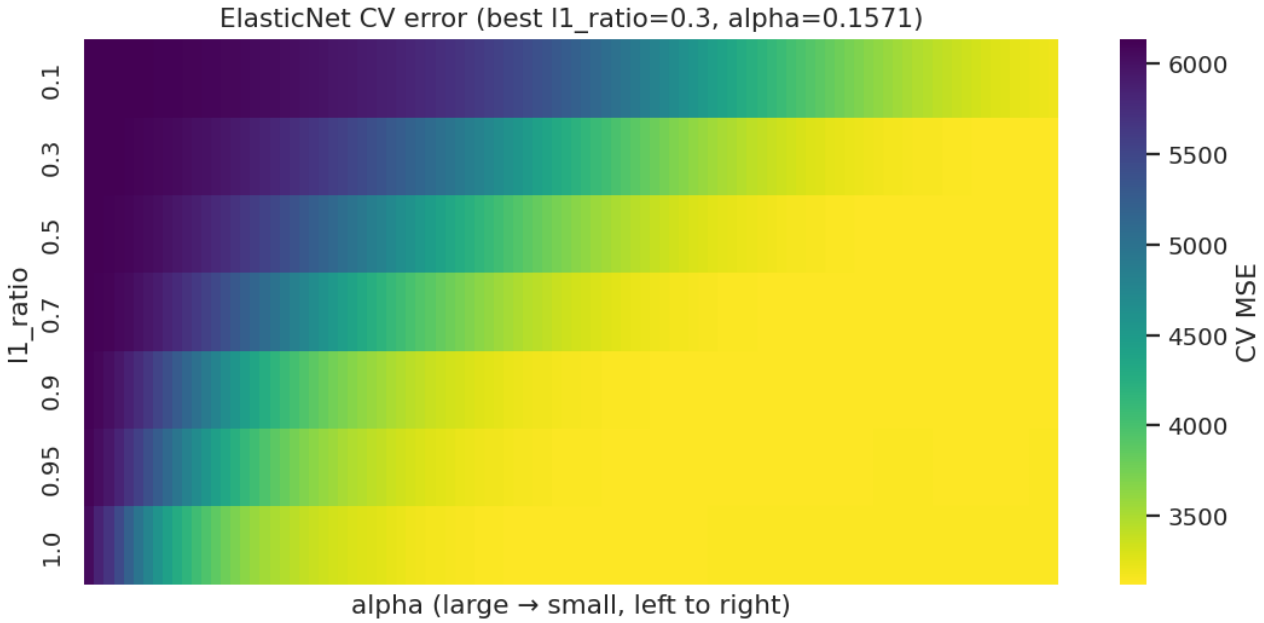
RMSE	MAE	R^2	Fit time (s)
52.92	42.79	0.471	0.047



5) ElasticNet

Diagnostic: CV error heatmap (alpha × l1_ratio)

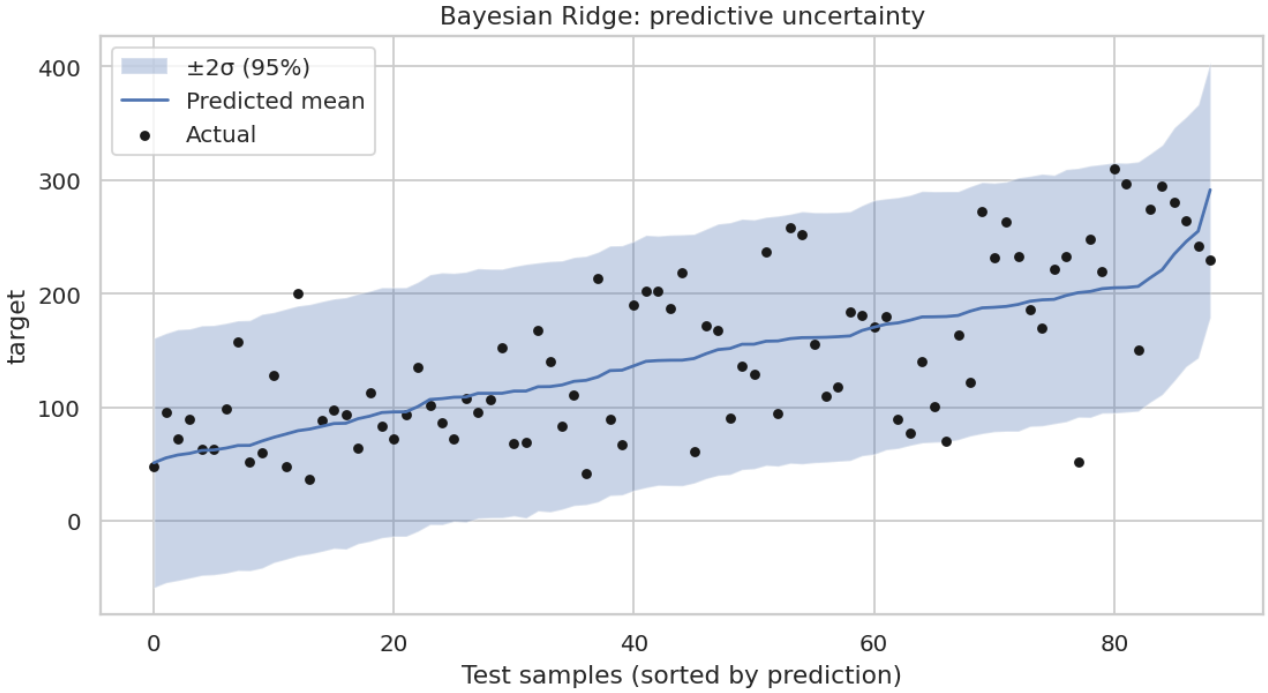
RMSE	MAE	R²	Fit time (s)
53.45	42.98	0.461	0.454



6) Bayesian Ridge

Diagnostic: predictions with uncertainty band

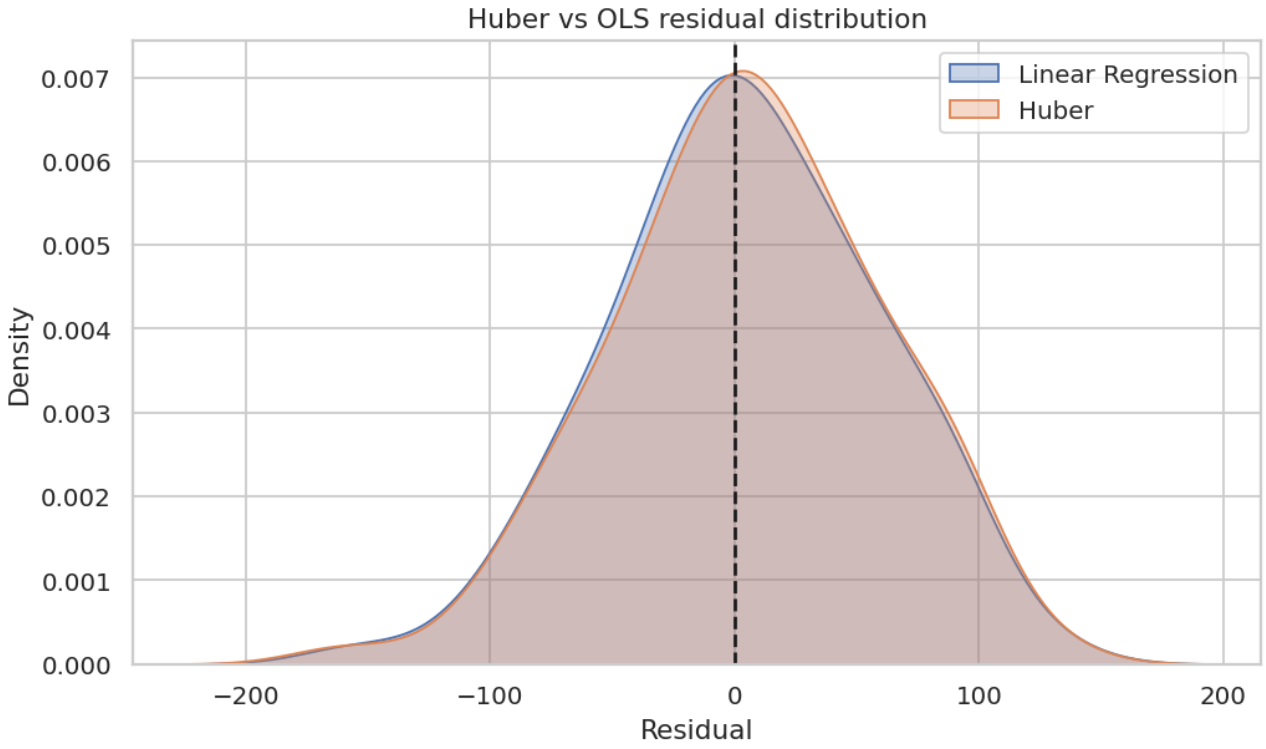
RMSE	MAE	R²	Fit time (s)
53.59	42.87	0.458	0.005



7) Huber Regressor

Diagnostic: outlier handling (residuals vs OLS)

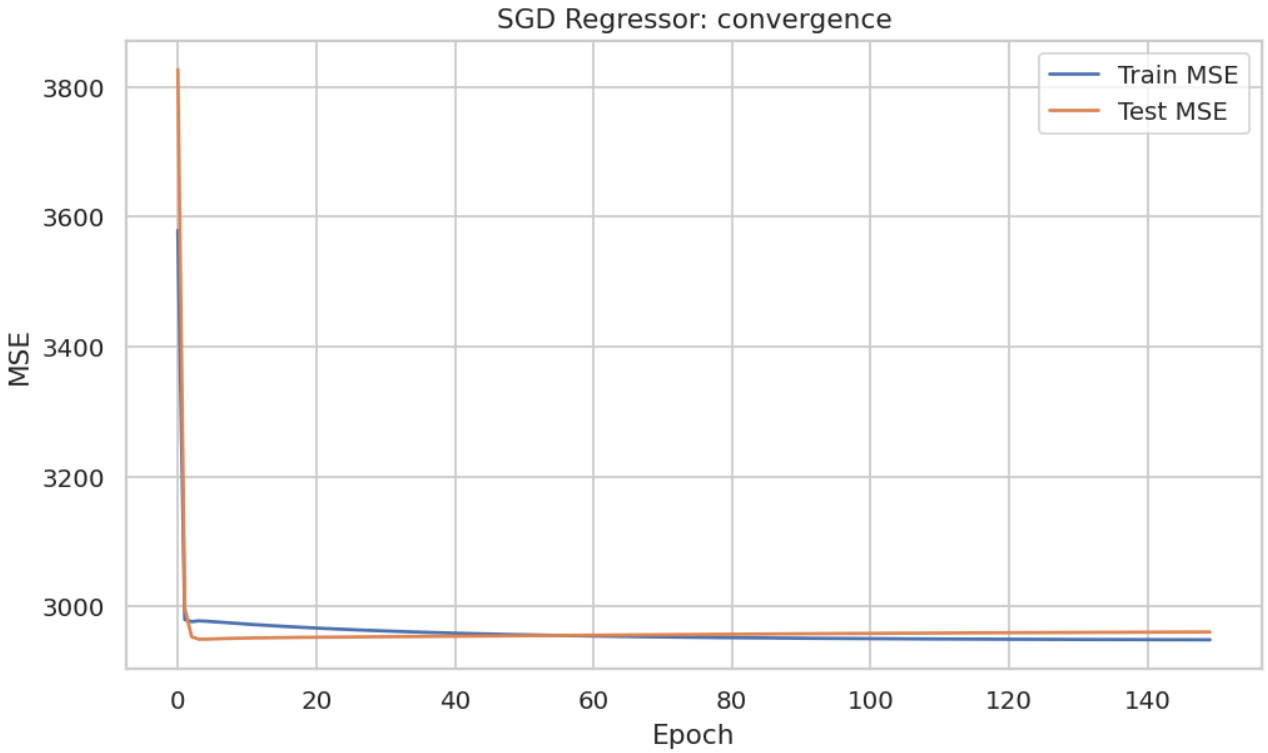
RMSE	MAE	R^2	Fit time (s)
54.14	42.88	0.447	0.007



8) SGD Regressor

Diagnostic: loss per epoch

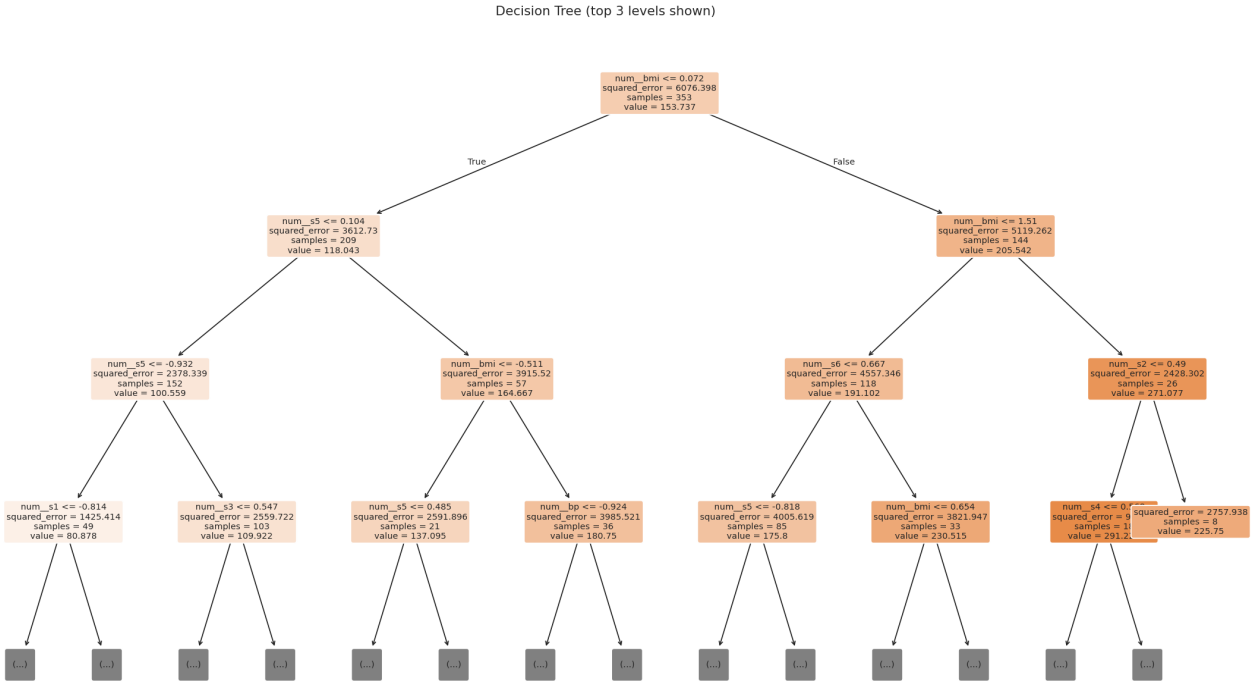
RMSE	MAE	R^2	Fit time (s)
54.41	43.28	0.441	—



9) Decision Tree

Diagnostic: the tree itself

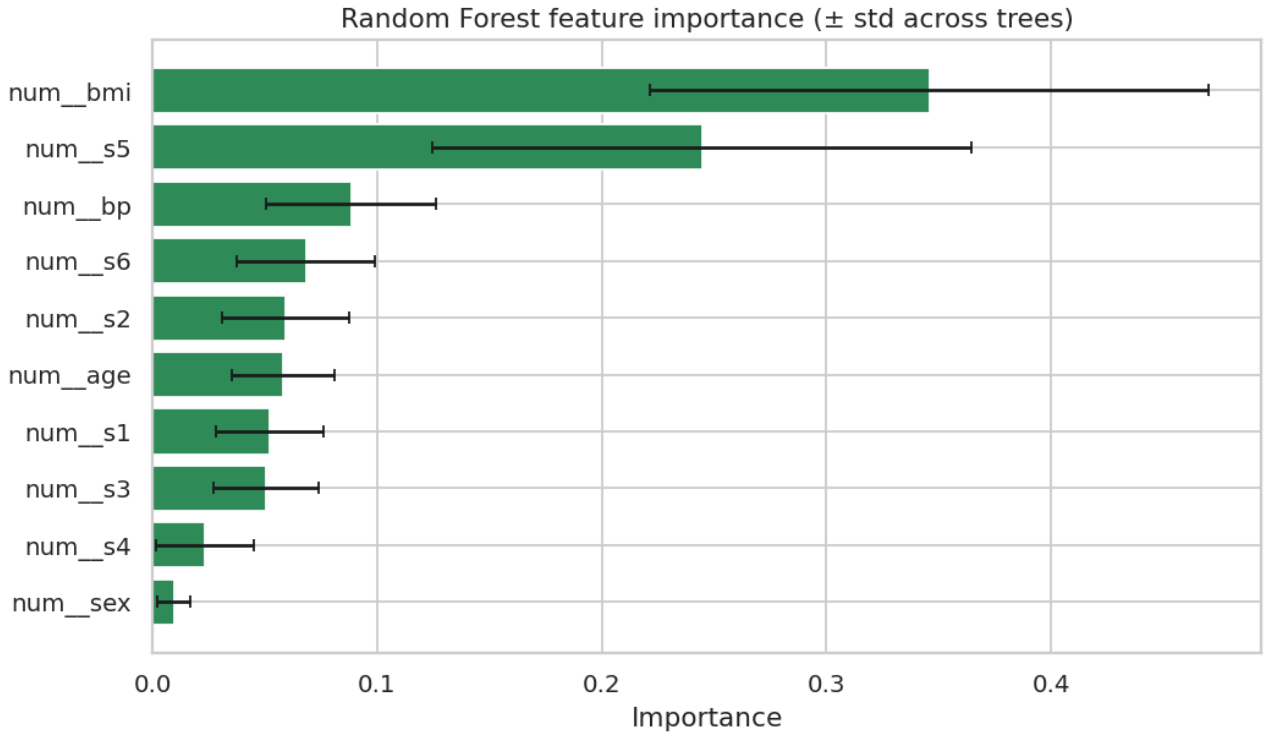
RMSE	MAE	R²	Fit time (s)
57.99	46.77	0.365	0.003



10) Random Forest

Diagnostic: feature importance with spread across trees

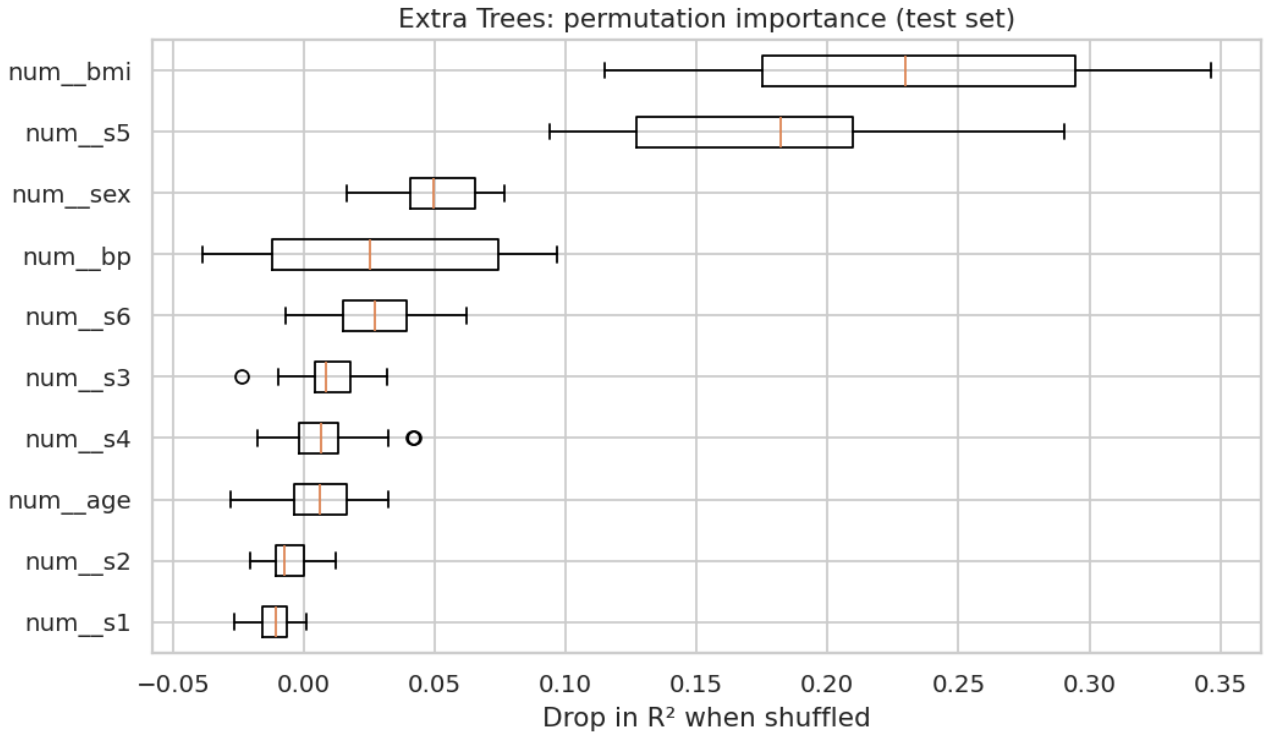
RMSE	MAE	R^2	Fit time (s)
54.92	44.75	0.431	0.641



11) Extra Trees

Diagnostic: permutation importance boxplot

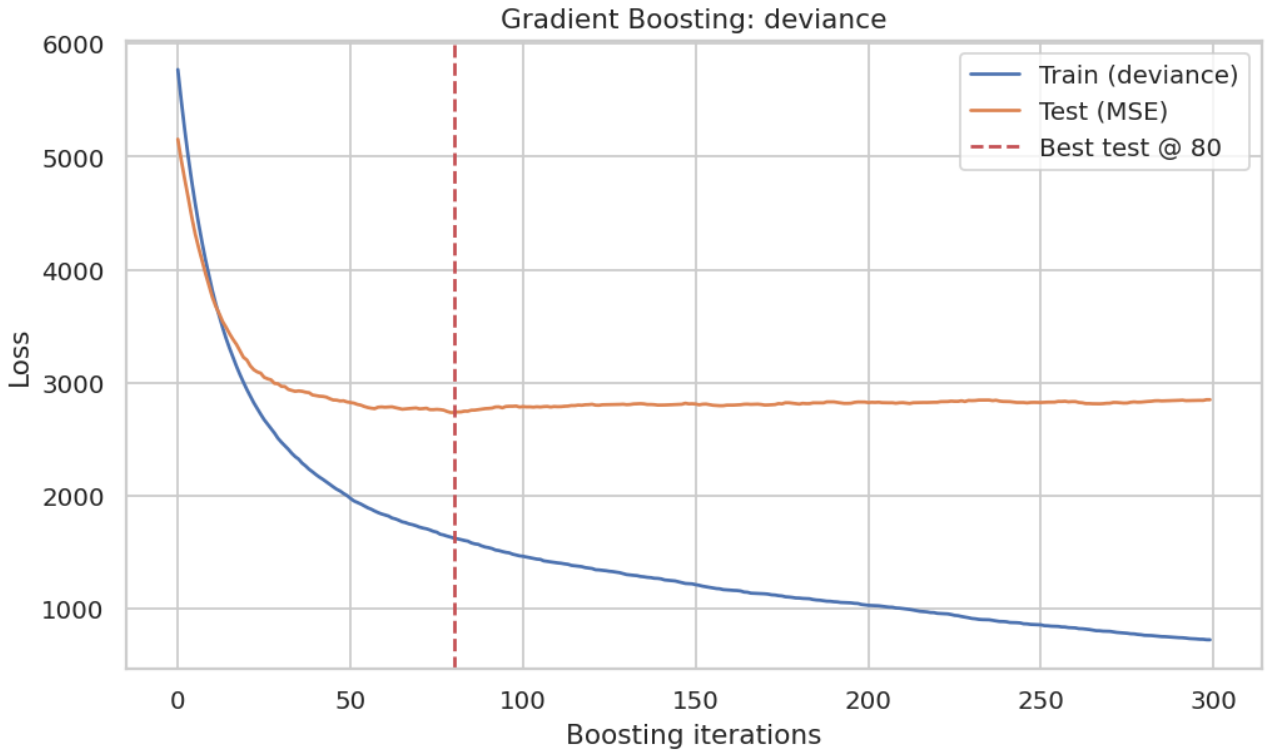
RMSE	MAE	R²	Fit time (s)
53.05	43.86	0.469	0.386



12) Gradient Boosting

Diagnostic: train vs test deviance by stage

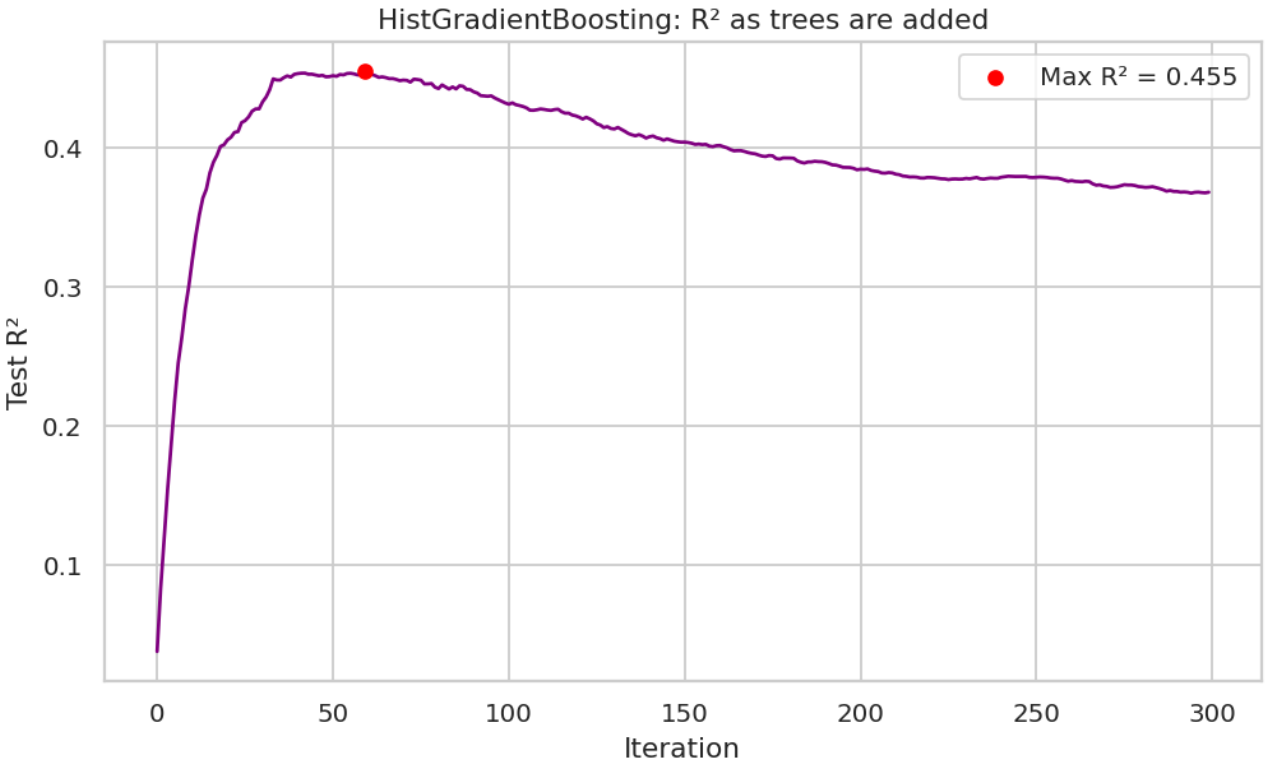
RMSE	MAE	R^2	Fit time (s)
53.39	43.62	0.462	0.361



13) Hist Gradient Boosting

Diagnostic: test R^2 by iteration

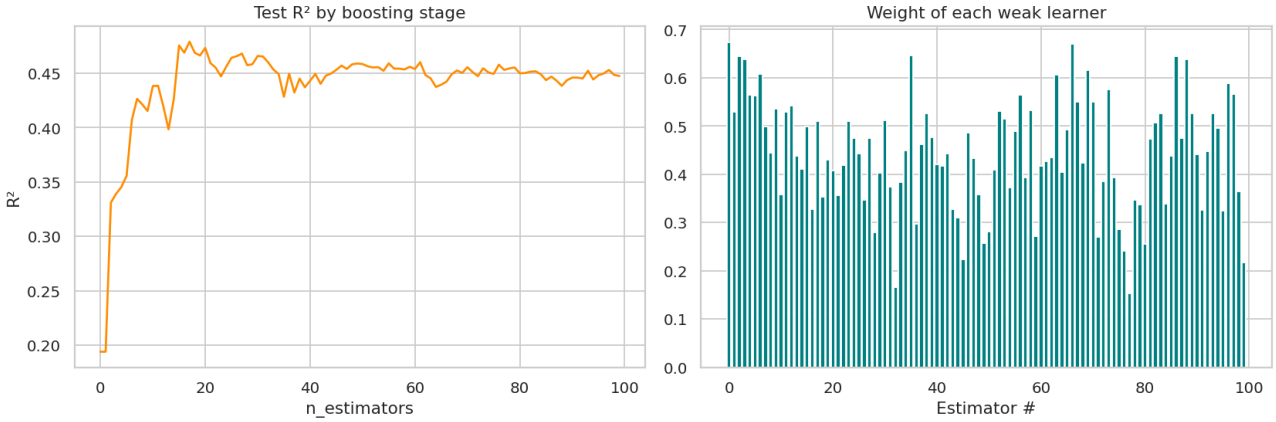
RMSE	MAE	R^2	Fit time (s)
57.87	46.09	0.368	0.233



14) AdaBoost

Diagnostic: staged R² and per-estimator weights

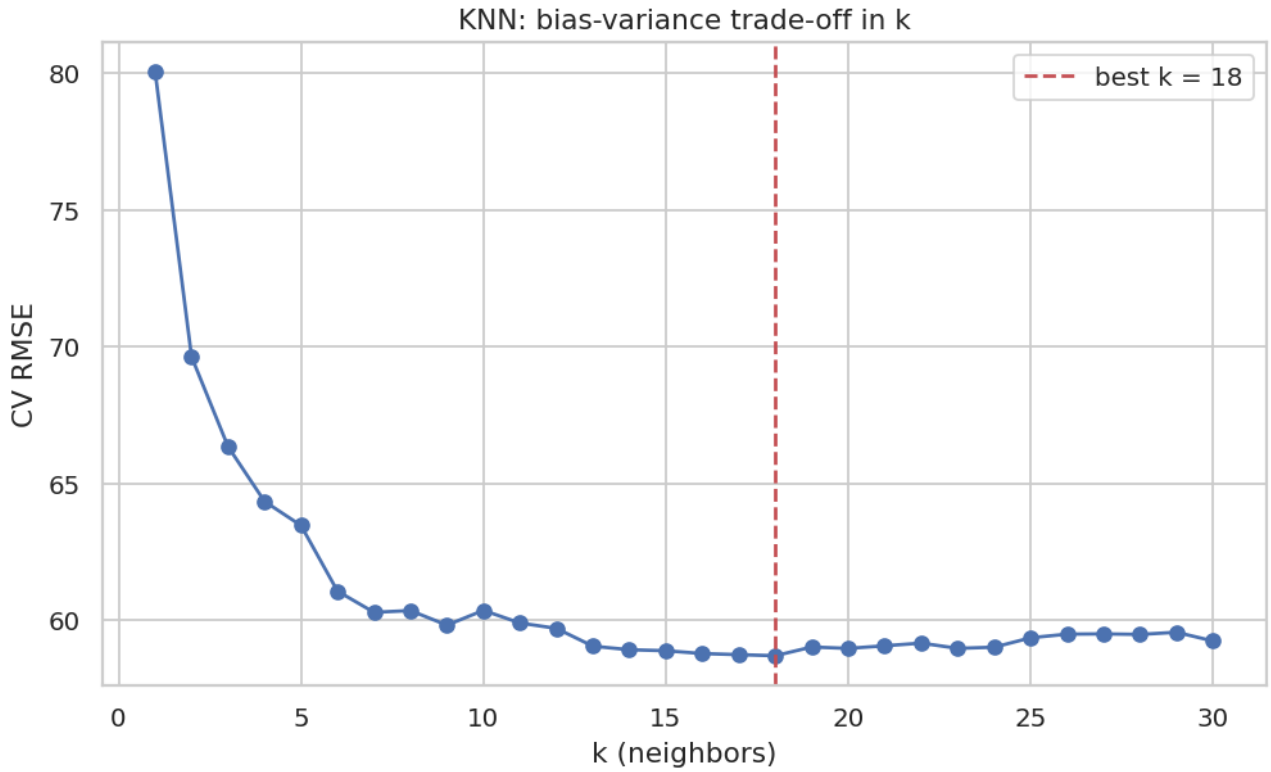
RMSE	MAE	R²	Fit time (s)
54.11	43.76	0.447	0.210



15) K-Nearest Neighbors

Diagnostic: CV error vs k

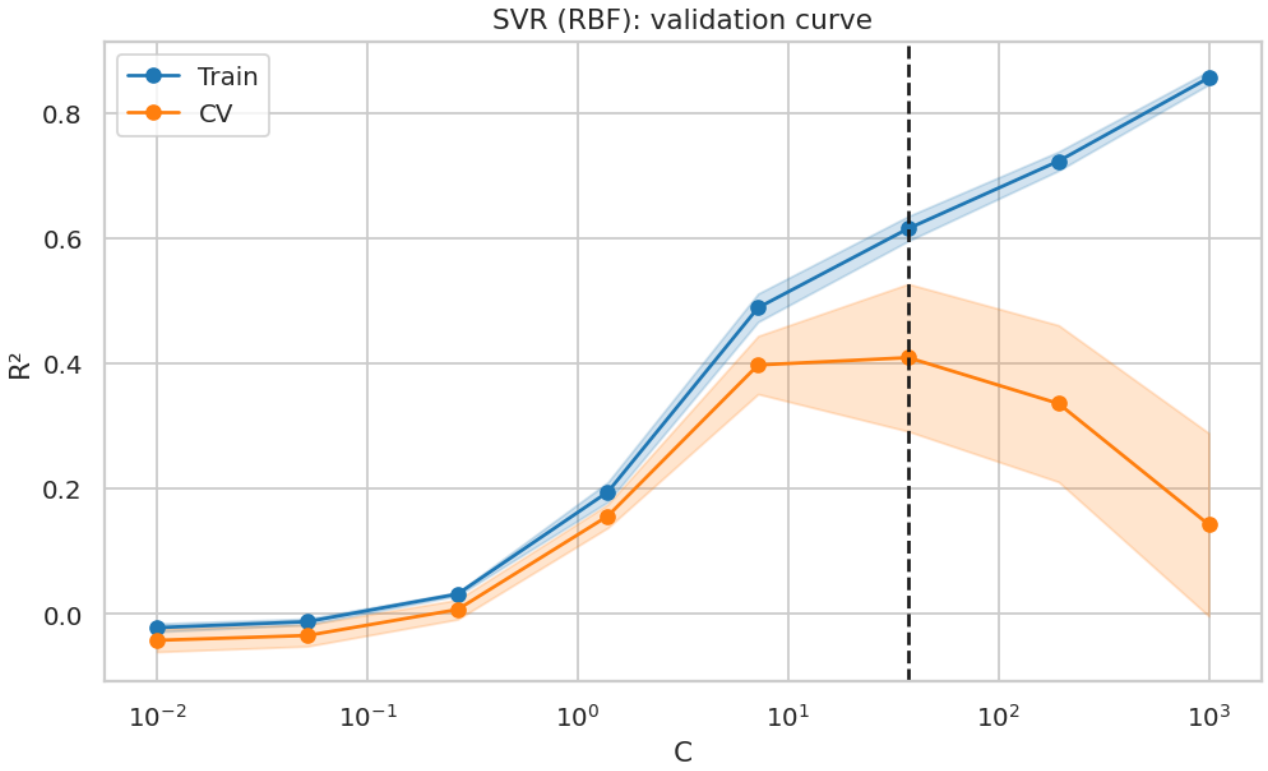
RMSE	MAE	R²	Fit time (s)
55.30	45.66	0.423	0.002



16) Support Vector Regression

Diagnostic: validation curve over C

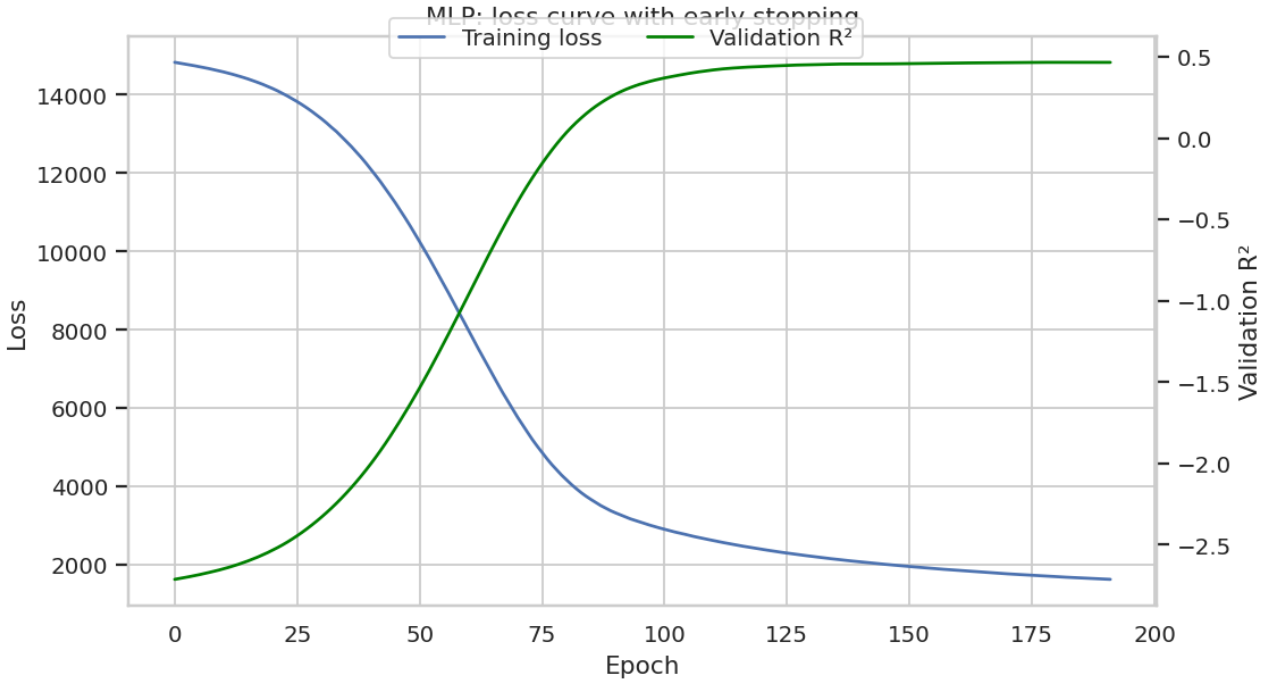
RMSE	MAE	R^2	Fit time (s)
50.10	39.10	0.526	0.009



17) MLP Neural Network

Diagnostic: training loss & validation score

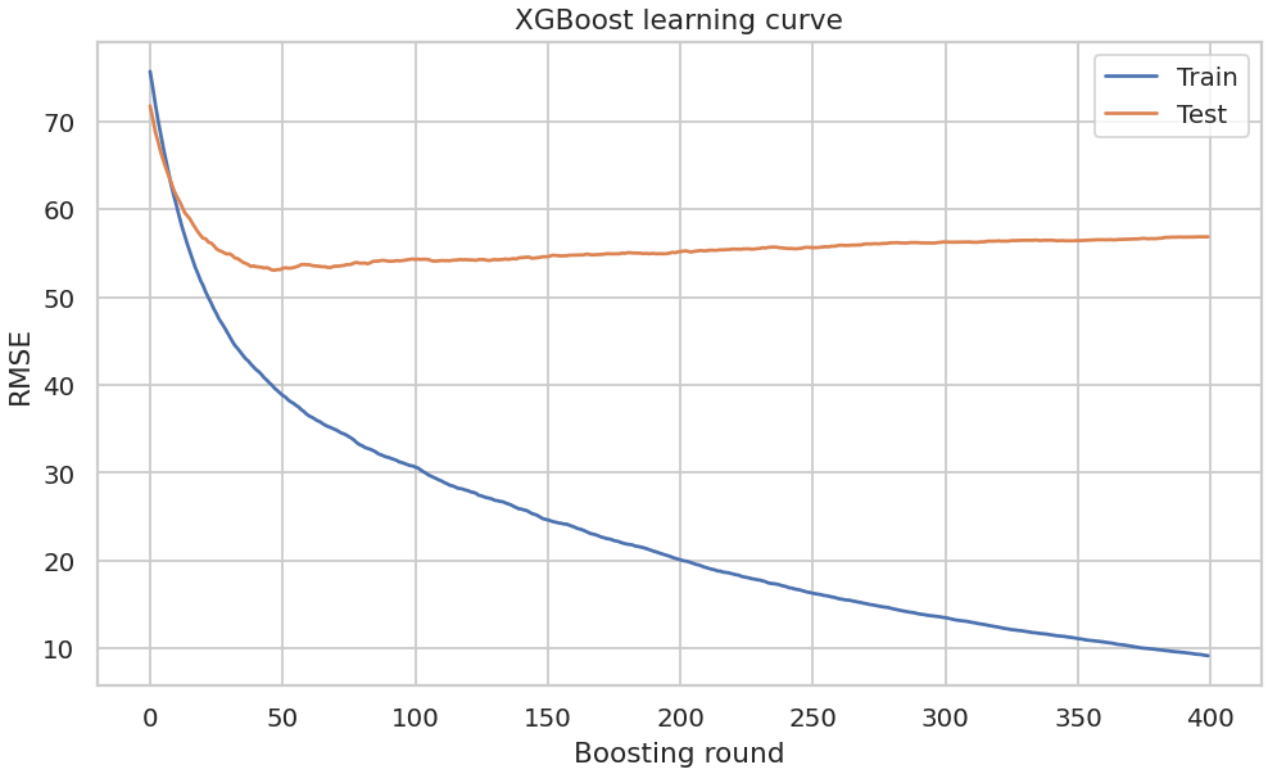
RMSE	MAE	R^2	Fit time (s)
60.47	48.16	0.310	0.303



18) XGBoost

Diagnostic: learning curve (RMSE per round)

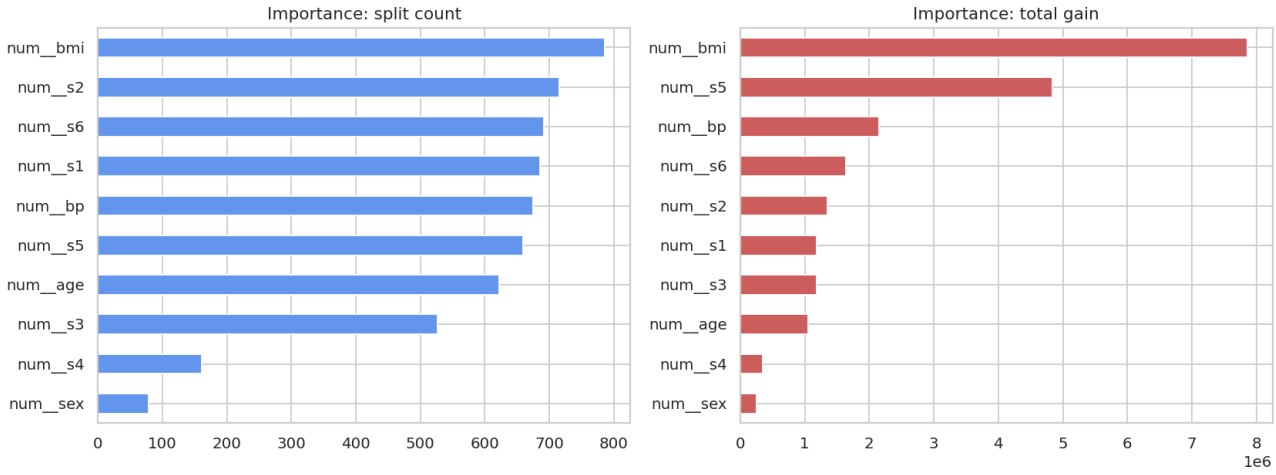
RMSE	MAE	R²	Fit time (s)
56.85	45.49	0.390	0.150



19) LightGBM

Diagnostic: feature importance by split count vs gain

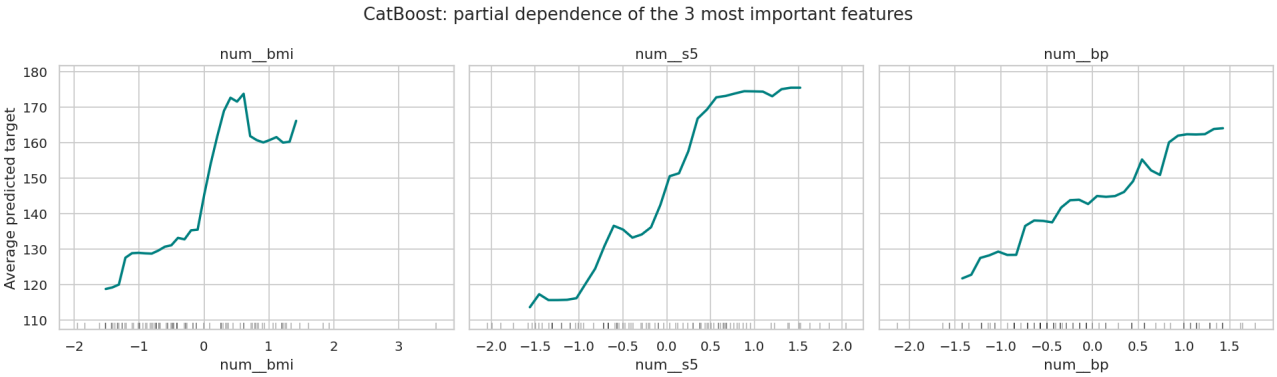
RMSE	MAE	R²	Fit time (s)
58.09	47.75	0.363	0.088



20) CatBoost

Diagnostic: partial dependence of the top features

RMSE	MAE	R²	Fit time (s)
52.58	42.61	0.478	0.495



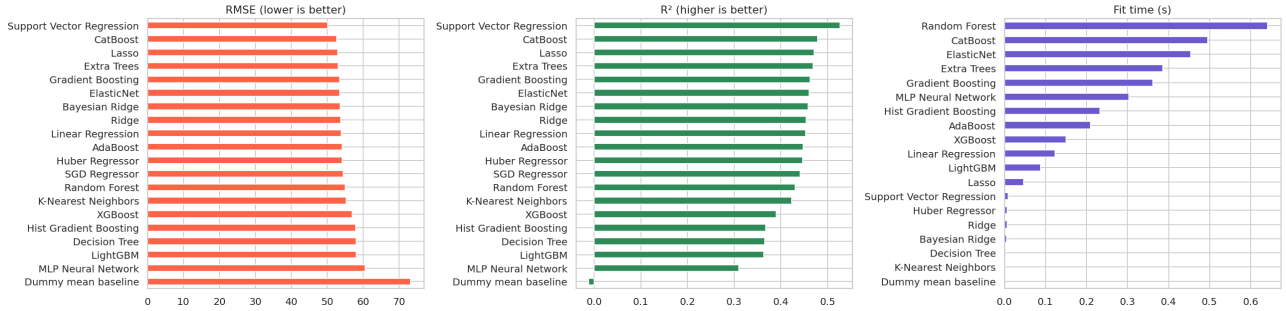
Model comparison dashboard

All 20 models ranked by test RMSE (lower is better); the dummy baseline is last.

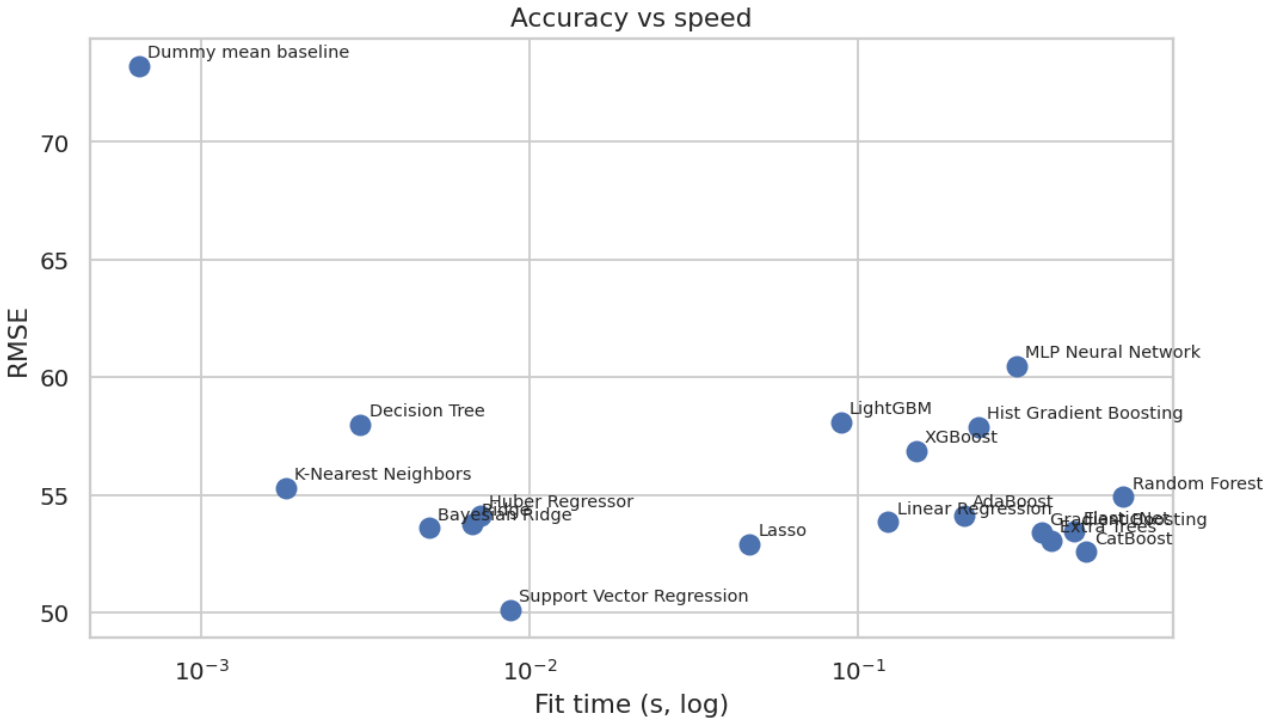
Rank	Model	RMSE	MAE	R^2	Fit time (s)
1	Support Vector Regression	50.10	39.10	0.526	0.009
2	CatBoost	52.58	42.61	0.478	0.495
3	Lasso	52.92	42.79	0.471	0.047
4	Extra Trees	53.05	43.86	0.469	0.386
5	Gradient Boosting	53.39	43.62	0.462	0.361
6	ElasticNet	53.45	42.98	0.461	0.454
7	Bayesian Ridge	53.59	42.87	0.458	0.005
8	Ridge	53.77	42.81	0.454	0.007
9	Linear Regression	53.85	42.79	0.453	0.123
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11	Huber Regressor	54.14	42.88	0.447	0.007
12	SGD Regressor	54.41	43.28	0.441	—
13	Random Forest	54.92	44.75	0.431	0.641
14	K-Nearest Neighbors	55.30	45.66	0.423	0.002
15	XGBoost	56.85	45.49	0.390	0.150
16	Hist Gradient Boosting	57.87	46.09	0.368	0.233
17	Decision Tree	57.99	46.77	0.365	0.003
18	LightGBM	58.09	47.75	0.363	0.088
19	MLP Neural Network	60.47	48.16	0.310	0.303
20	Dummy mean baseline	73.22	64.01	-0.012	0.001

Key takeaways. Support Vector Regression (RBF) is best on this small dataset (RMSE 50.10, R^2 0.526), followed by CatBoost (52.58) and Lasso (52.92). Regularised linear models perform on par with tree ensembles, which suggests the signal in this dataset is largely linear and the sample is small. The dummy mean baseline (RMSE 73.22, $R^2 \approx 0$) confirms every model learns real structure. The MLP (R^2 0.310), LightGBM (0.363), Hist Gradient Boosting (0.368) and Decision Tree (0.365) trail, consistent with overfitting or limited tuning on 353 training rows.

RMSE, R² and fit time across models



Accuracy vs speed (RMSE against fit time, log scale)



Actual vs predicted for all models

